E61: Exact Pattern-Formation Threshold in the Continuum

Mathematical proof for Manuscript 1

Abstract

For the centered Swift–Hohenberg linear evolution on $\mathbb{R}^d$, we prove that its spectral-instability threshold is exactly zero. We determine the spectrum and semigroup norm, distinguish strong stability from uniform exponential stability at criticality, characterize the unstable Fourier annulus, and derive the corresponding finite-periodic-domain threshold. The distinction between centered and uncentered control parameters is made explicit. The theorem concerns linear instability, not nonlinear pattern existence or localization.

1 Continuum theorem

Theorem 1 (E61: continuum spectral threshold)**.** *Let $d \geq 1$, $a > 0$, and $k_\star > 0$. On $L^2(\mathbb{R}^d)$ define*

$$L_r = rI - a(\Delta + k_\star^2)^2, \qquad D(L_r) = H^4(\mathbb{R}^d), \quad r \in \mathbb{R}. \tag{1}$$

Then

$$\sigma(L_r) = (-\infty, r], \qquad \|e^{tL_r}\|_{L^2 \to L^2} = e^{rt} \quad (t \geq 0). \tag{2}$$

Consequently the critical centered growth parameter is

$$\boxed{r_c = \min_{k \in \mathbb{R}^d} a(|k|^2 - k_\star^2)^2 = 0.} \tag{3}$$

More precisely:

1. *If $r < 0$, every L^2 solution decays exponentially.*
2. *If $r = 0$, every L^2 solution tends strongly to zero as $t \to \infty$, although the semigroup is not uniformly exponentially stable.*
3. *If $r > 0$, there exist nonzero L^2 data whose solutions grow exponentially in norm.*

The maximum Fourier growth rate is r, attained as a value of the symbol precisely on $|k| = k_\star$.

Proof. We use the unitary Fourier transform, so $\widehat{\Delta u}(k) = -|k|^2 \widehat{u}(k)$. Hence L_r is unitarily equivalent to multiplication by the real-valued function

$$\lambda_r(k) = r - a(|k|^2 - k_\star^2)^2. \tag{4}$$

For every $k \in \mathbb{R}^d$,

$$a(|k|^2 - k_\star^2)^2 \geq 0, \qquad \lambda_r(k) \leq r.$$

Since $k_\star > 0$, the sphere $S_{k_\star} = \{k \in \mathbb{R}^d : |k| = k_\star\}$ is nonempty, and $\lambda_r(k) = r$ for each $k \in S_{k_\star}$. It follows that

$$\sup_{k \in \mathbb{R}^d} \lambda_r(k) = r, \qquad \min_{k \in \mathbb{R}^d} a(|k|^2 - k_\star^2)^2 = 0. \tag{5}$$

The function λ_r is continuous, has range $(-\infty, r]$, and tends to $-\infty$ as $|k| \to \infty$. Its essential range is also $(-\infty, r]$: every neighborhood of a value in this range has a preimage of positive Lebesgue measure (including neighborhoods of the maximum r). The spectrum of an unbounded self-adjoint multiplication operator is its essential range, giving the first identity in (2). The Fourier formula for the semigroup is

$$\widehat{e^{tL_r}u_0}(k) = e^{t\lambda_r(k)}\widehat{u}_0(k), \qquad t \geq 0. \tag{6}$$

For $t \geq 0$, its operator norm is the essential supremum of the multiplier:

$$\|e^{tL_r}\|_{2\to 2} = \operatorname{ess\,sup}_{k\in\mathbb{R}^d} e^{t\lambda_r(k)} = e^{rt}.$$

In particular, if $r < 0$, then

$$\|e^{tL_r}u_0\|_2 \leq e^{rt}\|u_0\|_2 \longrightarrow 0 \qquad (t \to \infty).$$

At $r = 0$, Plancherel's identity gives

$$\|e^{tL_0}u_0\|_2^2 = \int_{\mathbb{R}^d} e^{-2at(|k|^2-k_\star^2)^2}|\widehat{u}_0(k)|^2\,dk. \tag{7}$$

The shell $S_{k_\star}$ has d-dimensional Lebesgue measure zero (including when $d = 1$, where it consists of two points). The multiplier in (7) therefore tends to zero for almost every k, while remaining bounded by 1. Since $|\widehat{u}_0|^2 \in L^1(\mathbb{R}^d)$, dominated convergence gives

$$\lim_{t\to\infty} \|e^{tL_0}u_0\|_2 = 0.$$

Nevertheless $\|e^{tL_0}\|_{2\to 2} = 1$ at every $t \geq 0$, so the semigroup is not uniformly exponentially stable. Note that the shell vectors are not nonzero L^2 eigenfunctions; the value zero is a spectral edge, not an L^2 point eigenvalue.

For $r > 0$, choose $k_0 \in S_{k_\star}$. By continuity of the symbol, there is a bounded open ball U centered at k_0 such that $\lambda_r(k) > r/2$ on U. Choose nonzero $u_0 \in L^2$ with $\widehat{u}_0$ supported in U. Then

$$\|e^{tL_r}u_0\|_2^2 = \int_U e^{2t\lambda_r(k)}|\widehat{u}_0(k)|^2\,dk \geq e^{rt}\|u_0\|_2^2,$$

and therefore $\|e^{tL_r}u_0\|_2 \geq e^{rt/2}\|u_0\|_2$. Thus instability first occurs as r crosses zero, proving (3) and the stated classification. □

2 Unstable spectral band and wavelength

Proposition 1 (Unstable annulus)**.** *For $r > 0$, the unstable wavevectors are exactly*

$$\{k \in \mathbb{R}^d : ||k|^2 - k_\star^2| < \sqrt{r/a}\}. \tag{8}$$

If $0 < r < ak_\star^4$, this is the annulus

$$\sqrt{k_\star^2 - \sqrt{r/a}} < |k| < \sqrt{k_\star^2 + \sqrt{r/a}}. \tag{9}$$

The fastest growth occurs at $|k| = k_\star$, corresponding to $\lambda_\star = 2\pi/k_\star$. As $r \downarrow 0$, the annulus collapses onto the shell $|k| = k_\star$.

Proof. The inequality $\lambda_r(k) > 0$ is equivalent to $a(|k|^2 - k_\star^2)^2 < r$, which is precisely (8). When $0 < r < ak_\star^4$, both bounds on $|k|^2$ are positive, so taking their square roots gives (9). The maximum is characterized in (5). A plane wave of wavenumber magnitude $k_\star$ has wavelength $2\pi/k_\star$. Both radii in (9) converge to $k_\star$ as $r \downarrow 0$. □

Remark 1 (Near-onset width)**.** *Writing $|k| = k_\star + \delta$ gives*

$$\lambda_r(k) = r - a(2k_\star\delta + \delta^2)^2 = r - 4ak_\star^2\delta^2 + O(\delta^3).$$

Thus the unstable half-width is asymptotic to $(2k_\star)^{-1}\sqrt{r/a}$ as $r \downarrow 0$.

3 Finite periodic domains

Proposition 2 (Periodic-box threshold)**.** *On the flat torus $\mathbb{T}_L^d = (\mathbb{R}/L\mathbb{Z})^d$, the linear growth rates are*

$$\lambda_n(r) = r - a\left(\left|\frac{2\pi n}{L}\right|^2 - k_\star^2\right)^2, \qquad n \in \mathbb{Z}^d.$$

The threshold is

$$r_c(L) = a \min_{n\in\mathbb{Z}^d}\left(\left|\frac{2\pi n}{L}\right|^2 - k_\star^2\right)^2 \geq 0. \tag{10}$$

Equality holds if and only if an admissible Fourier wavevector lies on $|k| = k_\star$. Moreover,

$$\lim_{L\to\infty} r_c(L) = 0. \tag{11}$$

Proof. The periodic Fourier basis has wavevectors $k_n = 2\pi n/L$, so each Fourier coefficient solves $\partial_t \widehat{u}_n = \lambda_n(r)\widehat{u}_n$. The first mode reaches nonnegative growth when r reaches the minimum in (10). The minimum exists because the nonnegative expression tends to infinity with $|n|$. It vanishes exactly when $|2\pi n/L| = k_\star$ for some $n \in \mathbb{Z}^d$.

For the limit, choose integers $m(L)$ satisfying $|m(L) - Lk_\star/(2\pi)| \leq 1/2$ and let $n(L) = (m(L), 0, \ldots, 0)$. Then

$$\frac{2\pi n(L)}{L} \longrightarrow k_\star e_1.$$

By the minimum property,

$$0 \leq r_c(L) \leq a\left(\left|\frac{2\pi n(L)}{L}\right|^2 - k_\star^2\right)^2 \longrightarrow 0.$$

The squeeze theorem proves (11). □

4 Centered versus uncentered parameters

Proposition 3 (Control-parameter translation)**.** *Suppose the original (uncentered) dispersion relation is*

$$\lambda(k) = r_0 + \alpha|k|^2 - b|k|^4, \qquad \alpha, b > 0.$$

Writing $k_\star^2 = \alpha/(2b)$ and $\mu = r_0 + \alpha^2/(4b)$ yields

$$\lambda(k) = \mu - b(|k|^2 - k_\star^2)^2. \tag{12}$$

Therefore the threshold is $\mu_c = 0$ in centered coordinates, and

$$\boxed{r_{0,c} = -\frac{\alpha^2}{4b}} \tag{13}$$

in the original coordinates.

Proof. Expanding the square in (12),

$$\mu - b\left(|k|^4 - 2k_\star^2|k|^2 + k_\star^4\right) = \mu - b|k|^4 + \alpha|k|^2 - \frac{\alpha^2}{4b} = r_0 + \alpha|k|^2 - b|k|^4.$$

Theorem 1 gives $\mu_c = 0$, which is equivalent to (13). □

5 Scope of the result

Theorem 1 proves a *linear spectral-instability threshold*. It does not assert nonlinear saturation, existence or uniqueness of patterns, orbital stability of nonzero equilibria, or localized finite-energy bound states. In particular, although an unprojected nonlinear Swift–Hohenberg evolution such as

$$\partial_t A = rA - a(\Delta + k_\star^2)^2 A - \beta|A|^2 A$$

has the same linearization at $A = 0$, the existence and stability of nonzero patterns require additional arguments. Nor does the present proof derive the linear operator from any more fundamental WCT model.